

Cardamom Annotation Guidelines

Project Description

The **Arabic Micro-Dialect Speech Data (Cardamom) Project** aims to build the first dataset representing fine-grained regional varieties of Arabic speech. Unlike existing corpora that operate at the country or major dialect level, **Cardamom** captures **micro-dialects** within key dialect groups, such as

Micro-Dialects

- Egyptian: Sa'idi, Port Saidi, Alexandrian, and Cairene
- Jordanian: Falahi, Madani, and Badawi
- Lebanon: Chamali, Jnoubi, B'albaki
- Mauritania: Northern, Southern, Western
- Saudi: Najdi, Hijazi, Eastern, Northern, and Southern
- Palestinian: Urban (Madani), Northern Rural (Fallahi), Southern Rural (Badawi)

Data Source

Speech data is collected primarily from **YouTube** sources (podcasts, interviews, series, etc). The collected material is then segmented into smaller utterances.

Objective

The primary objective of the **Cardamom Project** is to collect and annotate **speech data for each sub-dialect** of Arabic, in order to have a balanced and representative coverage. The annotation process is structured in two stages:

Stage 1: Core Annotation

- Transcription of spoken content
- Code-switching transcription (capturing language mixing)
- Speaker gender labeling
- Micro-dialect labeling

Potential Tasks

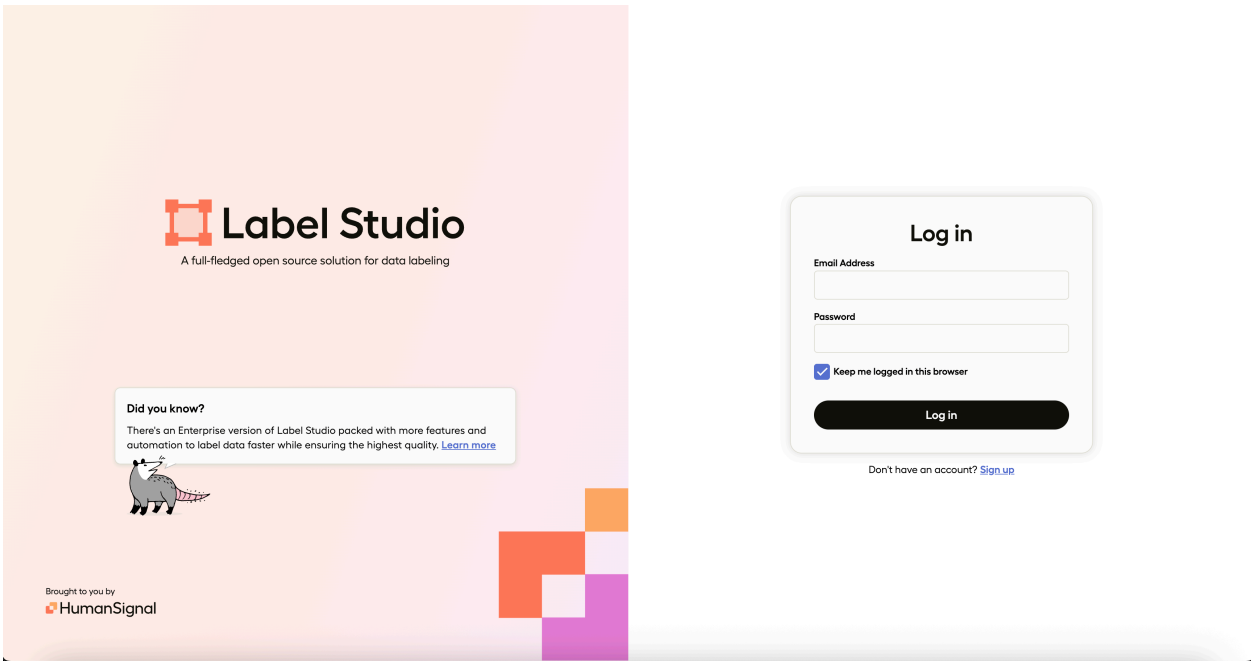
- **Automatic Speech Recognition (ASR)**
  - Arabic Level
  - Country Level
  - Micro-dialect Level
  - Code-switching
- **Arabic Dialect Identification (ADI)**
  - Country Level
  - Micro-dialect Level
- **Gender Identification**

Dataset Availability

The resulting dataset will be made **publicly available on Hugging Face** to support open research and community use.

Annotation Tool

For this project, we employed **Label Studio** (open-source platfrom) as our annotation platform.



To begin the annotation process, you must first create an account using the annotation tool.

How to sign up for the annotation tool?

1. Go to: [ANNONYMOUS LINK]
2. Click on `SIGN UP`. Add your email and password, then `CREATE ACCOUNT`

How to find your project

1. Once you log in to the annotation tool, you will find all the projects listed on the home page.
2. Enter your c**ountry** project, for example, **Egypt**, and you will find all the utterances there.

How to Access Your Project

1. After logging into the annotation tool, you will see a list of all available projects on the homepage.
2. Select the project for your **country** (e.g., **Egypt**) to view the utterances assigned there.

Conditions of Choosing Segment

What does “clean segment” mean?

- Each segment must contain only one speaker; no speaker overlaps are allowed.
- Speech within the segment should be audible and transcribable. Background noise is acceptable as long as the utterance remains understandable.
- Segments should preferably include at least three words and not exceed 30 seconds in duration.
- Ensure that the segment fully captures the speech from beginning to end, including every word and letter, without truncation. Pay close attention to the `start time` and `end time`.
- Avoid using purely non-dialectal utterances (e.g., full MSA sentences such as Quranic verses, hadeeth, or standalone greetings like "السلام عليكم"). It is acceptable if such expressions appear within a dialectal context (e.g., "السلام عليكم خيو شلونك").
- Avoid using sung phrases or segments delivered in a singing style.

Useful Writing Guidelines:

When writing Arabic words, avoid spelling errors, particularly with characters like hamza (ء), ta' (ة/ت), and ya' (ى/ي). Here are some useful materials:

- الفرق بين الياء والألف المقصورة
  - ref: https://www.almrsal.com/post/1194132

الفرق بين الياء والألف المقصورة هو

الياء	الألف المقصورة
الياء تنقط للتفريق بين الالف المقصوره وبينها	ترسم في اخر الفعل على صورة الياء
امثلة عليها: كتابي، قلمي، ذاتي، يعاني، الذي	امثلة عليها: رمى، سعى، ادعى، استوفى، اذا سبقت بياء رسمت بألف مثل أحيا، واستحيا، واذا كان الفعل ثلاثي مضارعه بالواو فترسم ألفًا مثل غزا ودعا
في الافعال مثل يبكي، يرثي	تكتب في اخر الاسم بصورة الياء مثل بشرى ومنتدى، واذا سبقت بياء تكتب رسمت ألفًا مثل دنيا وخطايا

- قواعد الهمزة

قواعد كتابة الهمزة في أول الكلمة

How to Transcribe Speech

When transcribing, please follow these guidelines:

1. Word Accuracy

- Check every single word in the transcription and correct any errors.
- Write the way you “write” in everyday conversations (e.g., on social media).
- **Discuss with your team and your team leader about how to** handle unique cases specific to your dialect.

2. Tanween and Diacritics

- For **tanween**, always place it **after the letter**, not before (to make preprocessing easier).
  - مثلاً ✅
  - مثلًا ❌
- Other diacritics and punctuation marks are **optional** (add them only if you are confident).

3. Numbers

- Always write numbers in **alphabetical format** rather than numerically.
  - ✅ انا عاوز عشرين بطاقة
  - ✅ قبل سنتين، يعني بالفين وعشرين، سافرت على تركيا

4. Code-Switching (Foreign Words)

When the speech includes foreign (non-Arabic) words:

1. First, transcribe the **entire utterance in Arabic** (including the foreign word, written phonetically in Arabic script).
   - Example: *ايش في بكرا بلانز*
2. Then, **copy-paste the same transcription** but replace the foreign word(s) with their **original form**.
   - Example: *ايش في بكرا plans*
3. If prefixes or suffixes are attached, separate them from the foreign word.
   - Examples:
     - *ها normalize بن*
     - *gym عال*
     - *ها analyze بن*

5. Laughter and Non-Speech Sounds

- If the speaker laughs, write it simply as: **هههه**
- No need to annotate other background noises

How to Label Micro-Dialect

- Each segment must be assigned **at least one micro-dialect** label.
- If the utterance contains elements from more than one micro-dialect, you may assign **multiple labels** (multi-label annotation).

- Example: *"ماشي ولا يهمك"* is a Jordanian utterance that could be spoken in both **Falahi** and **Madani** varieties. In this case, both labels should be applied.

FAQs

▼ Should I add the Tanween?

- Yes, please do. Preferably, add the tanween after the letter, not before. (This will make the preprocessing easier)
  - is correct مثلاً ;is wrong مثلًا

▼ How can I transcribe utterances that have foreign words (code-switching)?

- First, you have to write the full transcription in Arabic, including these words.
  - For example ايش في بكرا بلانز
- Second, you copy-paste the full Arabic transcription and just change the foreign word(s) into their original language (e.g., English)
  - For example: ايش في بكرا plans
- For cases that have prefixes/postfixes, write them separated from the foreign word:
  - For example:
    - بن normalize ها
    - عال gym
    - بن analyze ها

▼ How can I annotate segments that have only music/crying/screaming

- Just skip it.